

Unwind

Now that you've gone through the Zero1 Awards check out the Zero1Zero Awards as well

Community

Our keen readers have written about PHP and Information Security in this month's community

Digit 101 - Wireless connectivity

Wireless transmission isn't that tough to get your head around, there are plenty of abbreviations and numbers flying around and here is where you'll get the low down on all the mumbo-jumbo.

The days of using purely analog means to get the message across has long passed and we are now doodling with so much data that we've pretty much stuffed the airwaves with information. Right this very moment there are hundreds or maybe thousands of megabytes worth of data swirling around you. Let's have a look at some of the technology that goes behind all this.

Bluetooth

Bluetooth is a wireless data transfer protocol that is found on almost all mobile devices. 802.xx is the IEEE (Institute of Electronics and Electrical Engineers) protocol for wireless communication. 802.15 specifically stands for bluetooth. The range is significantly shorter compared to other protocols but the latest revision has helped increase both range and throughput speeds. Bluetooth is limited to just 8 active devices of which one is called the Master or the Server. There can be 255 more devices connected to the same network but only 8 may be active at any given time. As of Bluetooth 4.0 the protocol has started to use technology similar to the one in Wi-Fi to achieve higher speeds. The table below will elaborate further on the range and throughput speeds of bluetooth.

Bluetooth		
Version	Speeds	Range
1	721 kbit/s	10m
1.1	721 kbit/s	10m
1.2	721 kbit/s	10m
2	2.1 Mbit/s	10m
2.1	3 Mbit/s	10m
3	24 Mbit/s	100m
4	25 Mbit/s	100m

Wi-Fi

Everything has Wi-Fi in it, and the technology has entered the gigabit era as of 802.11ac standard. There are multiple standards under WiFi itself and there is a new one every other year. They enjoy increased speeds as they evolve. Ever since WiFi direct feature has made its way into the protocol, bluetooth has started losing favour. The following table lists the

Wi-Fi		
Version	Range	Speeds
802.11b	140m	11 Mbit/s
802.11a	120m	54 Mbit/s
802.11g	140m	54 Mbit/s
802.11n	250m	150 Mbit/s
802.11ac	250m	1300 Mbit/s

protocols and the speeds they offer along with the range. Do note that the range is depends on which band is currently in use.

Cellular					
Gen	Name	Full Form	Speeds	Type	Remarks
0G	Push To Talk, Mobile Telephone System, Improved Mobile Telephone System,	-	-	Analog	
1G	AMPS, NMT	Advanced Mobile Phone System, Nordic Mobile Telephone	-	Analog	
2G	GPRS	General Packet Radio Service	114 Kbps	Digital	
2.5G	EDGE	Enhanced Data GSM Evolution	384 Kbps	Digital	Transitional
3G	3G	Third Generation	3.1 Mbps	Digital	
3.5G	HSDPA	High-Speed Down-link Packet Access	14 Mbps	Digital	Transitional
3.75G	HSPA+	Evolved High-Speed Packet Access	168 Mbps	Digital	Transitional
4G	LTE	Long Term Evolution	299.6 Mbps	Digital	

Cellular

Cellular data communication was primarily analog when it came out and has migrated to digital since 2G (Second Generation). Currently 4G devices are getting more mainstream though only select regions in the country actually enjoy proper coverage. Data transfer via wireless medium came into being from 2G devices onwards where internet connectivity could be used alongside calling functionality. Here you have another table with all the specifications for cellular data transmission technology.

NFC

Near Field Communication is a relatively new protocol which works within a few inches only. It has been adopted as a means of payment authorisation (credit/debit cards and payment tokens for public transportation) and also as a digital visiting card. NFC is not meant for high throughput transmission. There are active and passive modes of transfer, while active has its own power source the passive mode uses the electro-

magnetic radiation from the active device to power itself.

RFID

Radio Frequency Identification is yet another technology where passive "tags" are used to identify objects and so on. Passive RFID tags are severely limited in range but active RFIDs can extend over kilometers. They are often used to track objects across

assembly lines, logistics and to track students around campus.

IrDA

The key difference between the remaining technologies and IrDA is that IrDA requires a

IrDA	
Revision	Speed
SIR	9.6-115.2 kbit/s
MIR	0.576-1.152 Mbit/s
FIR	4 Mbit/s
VFIR	16 Mbit/s
UFIR	96 Mbit/s
GigaIR	512 Mbit/s - 1Gbit/s
5/10GigaIR	5-10 Gbit/s

clear line of sight between the communicating devices in order for it to work. So this technology can be said to be a lot more secure and since the maximum range specified is one meter it all the more adds to the security. IrDA was popular in most phone models in the late 90's and early 2000's. Once again, we have a table highlighting the specifications across different versions. [\[1\]](#)